

Ashutosh Kumar

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Education

Indian Institute of Technology (IIT) Kanpur

2021-2025

B.Tech, Computer Science and Engineering, CPI: 8.74/10

Work Experience

Software Engineer

Jul'25-Present

Microsoft R&D India

- Built distributed **Spark** pipelines processing **real-time EventHub streams** through Synapse and Service Fabric into **Azure Data Explorer** for **large-scale analytics**
- Eliminated a heavy broadcast-join by a per-record computation over a purpose-built in-memory **data structure (indexed, binary-searchable buckets)**, achieving a **10% latency reduction** with lower executor memory and shuffle overhead
- Migrated the Spark pipelines from **HDInsight (Spark 3.3)** to **Azure Synapse (Spark 3.5)**, implementing **dual-runtime compilation** and validating output parity via side-by-side dual-sink execution on live production data
- Built a **multi-agent AI system** that analyzes production incidents end-to-end and recommends remediation, orchestrating isolated specialist sub-agents over **50+ metric queries** across 5 dashboards to **classify failure patterns** and **auto-generate RCA reports**, cutting **investigation time by 70%**
- Worked on a **Spark capacity-optimization** — profiling per-forest executor and core usage to flag pipelines handling low load with excess executors, validating executor cuts that reduced compute footprint while preserving throughput
- Reorganized applications into new **ADX workload groups** by **profiling request volume and CPU usage**, and tuned per-application request-rate limits to contain retry-heavy query traffic, protecting **overall cluster stability** and ensuring **fair resource sharing**

Quantitative Researcher Intern

May'24-Jul'24

Graviton Research Capital LLP

- Analyzed Nifty options to investigate the impact of **stop-loss** activity on price and identify potential trading opportunities
- Applied **quantile-binning** to engineered features to quantify the probability of order-book movement at integer price levels
- Built and backtested regression models (**Scikit-learn, XGBoost, TensorFlow**) on the identified indicators, improving return-prediction accuracy over baseline

Projects

AeroDrome Analysis

Feb'25 - Apr'25

- Implemented **linear-time conflict-serializability** checking in **C++** using vector clocks for **transactional memory**
- Used **Intel Pin** to instrument memory accesses at the instruction level, attributing each to its transaction
- Enabled **on-the-fly atomicity checking**, detecting violations during execution rather than post-hoc log analysis

Python Compiler

Jan'24 - Apr'24

- Developed a custom compiler that translates Python code into **x86 assembly code** that can be compiled using **gcc**
- Utilized **Flex** and **Bison** to implement the lexer and parser, ensuring accurate token generation and syntax analysis
- Supported various key Python features, including **classes, functions, arrays, and strings**, recursive function calls, inheritance and constructors

GemOS

Aug'23 - Nov'23

- Built support for **strace** and user space function call tracing functionality by implementing syscalls in teaching OS
- Implemented memory management methods **mmap, munmap** and **mprotect** and syscalls **memalloc** and **memfree**
- Developed page-fault handler for lazy allocation and **copy-on-write** fault handler for efficient memory management on fork

Zero Shot Machine Unlearning

Jan'24 - Apr'24

- Gained understanding on methods of **Error Minimisation-Maximisation Noise** and **Gated Knowledge Transfer**
- Added an **entropy filter** to the generator in the **GKT model** to enhance the quality of images produced for retraining achieving an accuracy of 98.3% at significantly reduced number of epochs thus reducing the retraining time
- Replaced the generator with a **Deep Inversion** model to produce a more accurate and high-fidelity dataset for retraining

Technical Skills

- **Programming Languages:** C/C++, C#, Scala, Python, Java
- **Libraries & Utilities:** Linux, Numpy, Pandas, Git/Github, Bash, Spark, Tensorflow, XGBoost
- **Cloud Platforms:** Eventhubs, Azure Data Explorer, Service Fabric Cluster, Synapse, ADLS

Academic Achievements

- Conferred with **Academic Excellence Award (2022)**, for outstanding performance in academics, by IIT Kanpur
- Secured **All India Rank 1039** in **JEE Advanced 2021** among 200K candidates
- Secured **All India Rank 2077** in **JEE Mains 2021** among 1 million candidates
- Secured an **All India Rank of 1531** in Kishore Vaigyanik Protsahan Yojana (**KVPY-SX 2020**), IISc
- Qualified for **Indian National Mathematics Olympiad(INMO)**, among the top 100 students in the country
- Awarded **National Talent Search Examination(NTSE)**, **2019** scholarship, by NCERT, Govt. of India
- Qualified for **Indian National Astronomy Olympiad(INAO)** 2021 conducted by HBCSE, TIFR
- Qualified for **Indian National Physics Olympiad(INPhO)** 2021 conducted by HBCSE, TIFR